

CS3110- Introduction to Computer Graphics

Lab Assignment 2

Fall 2026

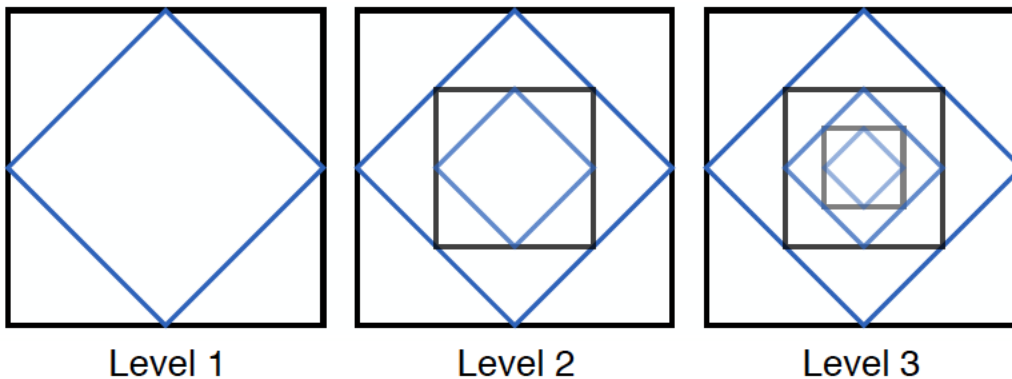
Dr. Ubaid Abbasi

Objective

The purpose of this assignment in CS3110 is to provide you opportunities to further familiarize yourself with WebGL programming environment by drawing simple, yet interesting two-dimensional geometry. The bulk of your work will be to write code in WebGL to generate the correct geometry for the polygonal, parametric, and fractal shapes you're asked to draw. After drawing these shapes you are also required to perform basic transformation such as Scaling, Translation, rotation and animating these geometric patterns.

Task I: Squares & Diamonds (Basic Transformation)

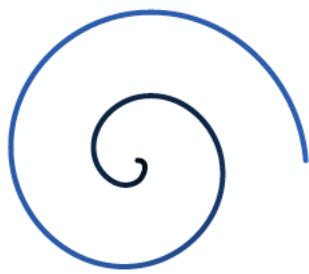
Create a program, or modify the provided template, to draw a pattern of nested squares and diamonds as shown in the diagram below. Choose different/same colors for the squares and diamonds. Use keyboard input to provide a means for interactively changing the levels of nesting drawn (i.e. Number of squares/diamonds). Allow for at least 6 levels of nesting. After drawing provide control for continuously rotating the nested squares around the Z-axis.



Task II: Parametric Spiral (Basic Transformation)

Add the ability to draw a second scene, a spiral, to your program from Part I. Construct your spiral with a color gradient so that the color of the curve at the center of your window smoothly transitions to a different color at the edge as you trace the curve's path, as shown below. Provide a means for interactively changing the number of revolutions the spiral makes as it goes from the

center to the edge of the window. After drawing provide control for continuously rotating the nested squares around the Y-axis.



2 Revolutions



3 Revolutions



4 Revolutions

Task III: Parametric Spiral (Basic Transformation)

Add a third scene to your program to draw a Sierpinski triangle or a two-dimensional Menger sponge. (You may do both if you'd like.). Both are well-known fractal shapes with very interesting mathematical properties.

Start with an equilateral triangle for the Sierpinski triangle and centre it in your window. The next iteration of the Sierpinski triangle is generated by splitting the triangle into three smaller, equally-sized triangles, with side lengths half of that of the original, and placing them within the space of the original so that their corners just touch, as shown below. Repeat this process for all triangles in the fractal to generate additional levels. After drawing provide control for continuously rotating the nested squares around the X-axis.



Base Triangle

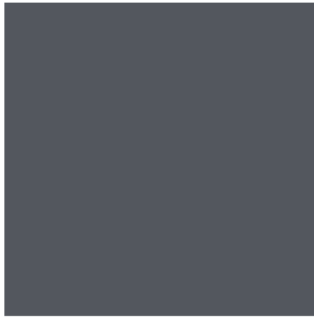


1 Iteration



2 Iterations

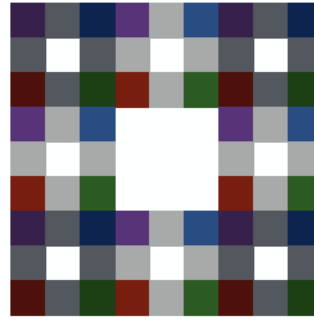
The Menger sponge is constructed in a similar fashion, but by starting with a square. The next iteration is generated by splitting the square into eight smaller squares, each with sides one-third the length of the original, and placing them on the inner perimeter of the original. The result should look as though a single small square were removed from the centre. Again, additional levels of the fractal are generated by repeating this process.



Base Square



1 Iteration



2 Iterations

Deliverables

1. Your code + any supporting files (for instance, if you put custom functions or points arrays in separate .js files)
2. An image of your final product